

Lifetime Performance and Degradation of Floating Photovoltaic Systems Under Humid and Saline Conditions

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Abstract—This study develops a coupled hourly simulation framework to quantify the competing thermal, humidity, soiling, potential-induced degradation (PID) and chloride-corrosion effects on floating photovoltaic (FPV) systems in hot, humid environments. A single over-water microclimate drives module temperature, module-surface humidity and dew, a dust and sea-salt soiling balance with NaCl deliquescence, and a four-mechanism module degradation model. The mechanisms are intrinsic ageing, Peck temperature–humidity damage with lagged encapsulant moisture, PID with salt-assisted surface leakage, and corrosion scaled by time of wetness and chloride deposition. For a representative Visakhapatnam (India) climate, the model compares coastal land PV, freshwater FPV and saline FPV. Under the baseline assumptions, freshwater FPV yields +1.6% in year 1 and +2.1% over 25 years. Saline FPV yields −1.1% and −4.3%, with a predicted degradation rate of 1.16%/yr against 0.88%/yr on land. A 1,000-sample Monte Carlo over 26 uncertain parameters separates parameter uncertainty from weather variability. The saline deficit is robust: the 95% interval of its lifetime-energy difference is [−11.2, 0.7]%. The freshwater advantage is not: [−3.6, 6.4]%, positive in 73% of samples, and controlled mainly by the FPV heat-loss coefficients and the dust load at the land alternative. Freshwater FPV loses its lifetime advantage if the dew point over water rises by more than 2.7 K. Mitigation is modelled with explicitly assumed reduction factors, not inferred from certification. These results are model-based scenario predictions that still require field validation.

Index Terms—Floating photovoltaics, humidity, salinity, soiling, potential-induced degradation, corrosion, module temperature, uncertainty quantification, degradation rate, levelized cost of energy.

NOMENCLATURE

G	Plane-of-array (POA) irradiance (W/m ²).
T_a, T_d, T_m	Air, dew-point and module temperature (°C).
v	Wind speed (m/s).
U_0, U_1	Faiman heat-loss coefficients.
RH_s, RH_e	Module-surface and encapsulant relative humidity (%).
m_d, m_s	Dust and salt surface density (g/m ²).
SR	Soiling ratio.
TOW	Time of wetness (h/yr).
DCl	Chloride deposition rate (mg/(m ² d)).
R	Annual module power degradation rate (%/yr).

μ_{PID}, μ_C	Assumed mitigation factors for PID and corrosion.
E_a	Activation energy (eV).
k_B	Boltzmann constant (8.617×10^{-5} eV/K).
PR	Performance ratio.

I. INTRODUCTION

FLOATING photovoltaic (FPV) systems place PV arrays on pontoons over reservoirs, lakes, lagoons and sheltered coastal waters. They avoid competition for land, reduce evaporation from water-supply reservoirs and can share grid infrastructure with hydropower plants [1]–[5]. India has many reservoirs, salt pans and backwaters along a coast with high irradiation, so FPV is attractive there, but many candidate sites are hot, humid and exposed to marine aerosol [6].

The performance case for FPV usually rests on the cooling effect of the water. In tropical Singapore, well-ventilated FPV designs ran cooler and reached higher performance ratios than a rooftop reference [7], with irradiance-weighted temperatures up to 14.5 °C lower [8]. Offshore simulations also predict higher yields than on land [9]. More recent measurements show that the cooling effect varies widely with design. Modules that are not in thermal contact with the water behave much like land-based modules [10]. Measured heat-loss coefficients span 22.6–33.7 W/(m² K) for several air-cooled systems [11], [12], and wind-dependent terms of 2.7–5.1 W s/(m³ K) have been reported [13]. A multi-year, humid-subtropical dataset found the FPV array only about 0.8 K cooler than a co-located land reference [14].

The water body also raises humidity and, at saline sites, airborne salt. The Peck model describes humidity-driven damage as an Arrhenius function of temperature times a power of relative humidity [15], and moisture enters polymer-backsheet modules over hours to days [16]. Combined-stressor lifetime models confirm that humidity is a first-order variable [17]. PID is strongly accelerated by humidity [18]–[22]. Sea salt lowers the surface resistance of module glass by several orders of magnitude as RH rises [23], and salt raises leakage current [24]. Field surveys report higher degradation in hot climates [25], and in India the hot–humid zones are among the worst affected [26]. Chloride deposition and time of wetness define atmospheric corrosivity [27], and salt mist is a module qualification stress [28]. Salt crystallization dominates the deposits on coastal FPV modules [29], and salt exposure acts on PV through soiling, corrosion, leakage-driven PID and barrier ageing [30].

Manuscript prepared October 2026; revised after peer review.

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The simulation code, input parameters, random seeds and all scripts used to produce the results are openly available (see the Data Availability section).

TABLE I
SCOPE OF REPRESENTATIVE STUDIES (✓ = ADDRESSED)

Study	FPV	Therm.	Hum.	Soil.	Salt	PID	Corr.	Life	LCOE
Liu <i>et al.</i> [7]	✓	✓							
Dörenkämper <i>et al.</i> [8]	✓	✓							
Lindholm <i>et al.</i> [10]	✓	✓							
Golroodbari, van Sark [9]	✓	✓							
Ahmad <i>et al.</i> [29]	✓	✓		✓	✓				
Kaaya <i>et al.</i> [17]	✓	✓	✓					✓	
Mannino <i>et al.</i> [31]	✓	✓	✓					✓	
Ajmal <i>et al.</i> [32]	✓	✓	✓		✓			✓	
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓

Two recent studies come closest to this work. Mannino *et al.* [31] applied existing degradation models to onshore and offshore meteorological data. Ajmal *et al.* [32] added salinity to climatic degradation models for offshore modules and calibrated them by optimization. Table I compares the scope of representative studies. To the author's knowledge, no study yet resolves, within one consistent microclimate, both the yield benefits of FPV and the humidity-, salt- and PID-driven penalties, and compares freshwater with saline sites over the plant lifetime and in cost terms.

This paper makes the following contributions:

- 1) A coupled hourly model chain in which one over-water microclimate drives module temperature, dew, spectral response, soiling and four module-degradation mechanisms, so that their trade-offs are resolved consistently.
- 2) A moisture formulation based on module-surface RH with a lagged encapsulant response. It captures why cooler modules can see more moisture, and it couples PID to salt-assisted surface leakage.
- 3) A comparison of coastal land PV, freshwater FPV and saline FPV covering year-1 losses, mechanism-resolved degradation, 25-year energy and LCOE. Mitigation options are evaluated one by one, with explicitly assumed effects.
- 4) An uncertainty analysis that separates weather variability (20 synthetic years) from parameter uncertainty (a 1,000-sample Monte Carlo over 26 parameters). It is complemented by break-even thresholds for salt deposition, chloride deposition and dew-point rise, and by economic break-even values for CAPEX, land cost and water value.

Scope. The degradation results describe the *electrical degradation of the PV modules*. Degradation of the floating system itself (fatigue of floats and moorings, corrosion of structures [33], [34], storm damage, bio-fouling) is outside the model, so the lifetime energy reported here is not a complete FPV lifetime prediction.

II. SITE AND STUDY SCENARIOS

The reference site is Visakhapatnam (17.69° N, 83.22° E) on the Bay of Bengal. Its climate is tropical, with a hot pre-monsoon season (April–May), the south-west monsoon (June–September) and the north-east monsoon (October–November). An hourly typical year was synthesized from monthly normals (Section III-A). It has a GHI of 1,889 kWh/m², a mean air

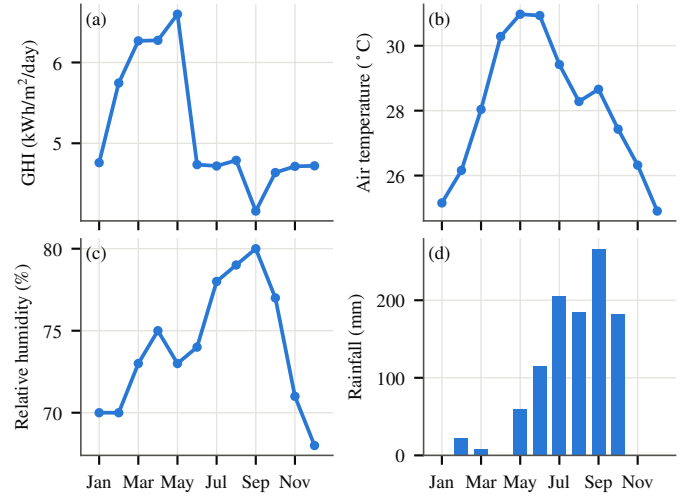


Fig. 1. Monthly climate of the synthetic Visakhapatnam typical year: (a) daily GHI, (b) air temperature, (c) relative humidity and (d) rainfall.

TABLE II
BASELINE SCENARIO PARAMETERS (ASSUMPTIONS)

Parameter	Land	FPV fresh	FPV saline
Tilt (°)	15	12	12
Albedo	0.20	0.07	0.07
U_0 (W/(m ² K))	25.0	35.0	35.0
U_1 (W s/(m ³ K))	6.84	8.0	8.0
Air–water coupling α	0	0.25	0.30
Dew-point rise ΔT_d (K)	0	1.0	1.5
Wind factor k_v	1.00	1.15	1.25
Dust deposition (g/(m ² d))	0.15	0.04	0.03
Salt deposition (g/(m ² d))	0.015	0.010	0.060
D_{Cl} (mg/(m ² d))	30	8	180
Wiring loss (%)	1.5	2.0	2.0
Wave-motion mismatch (%)	0	0.7	1.0
Availability (%)	99.0	98.5	98.0
CAPEX (INR/Wp)	35	45	52
O&M (INR/kWp/yr)	450	550	750

temperature of 28.1 °C, a mean RH of 74.0% and 1,043 mm of rain (Fig. 1).

Three 1-MWp (DC) plants with a DC/AC ratio of 1.2 and monocrystalline PERC modules (−0.35%/K) are compared (Table II). *Land PV* is a coastal ground-mounted plant on open racks with 15° tilt. *FPV freshwater* is a pontoon plant with 12° tilt on an inland reservoir with low chloride deposition. *FPV saline* uses the same design on near-shore or brackish water (lagoon, salt pan, harbour) with high salt and chloride loading. The baseline values are scenario assumptions informed by the literature, not site measurements; their uncertainty is propagated in Section III-I.

III. MODELING FRAMEWORK

The model chain is shown in Fig. 2. All sub-models run hourly over one year; degradation rates computed from the hourly stresses are projected over a 25-year life. The established irradiance and electrical models follow their reference formulations, as implemented, for example, in pvlib [35].

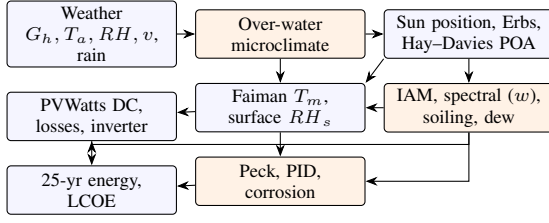


Fig. 2. Coupled model chain. Shaded blocks carry an explicit dependence on humidity or salinity.

TABLE III
SYNTHETIC WEATHER VERSUS INPUT CLIMATOLOGY (20 YEARS)

Variable	Normal	Generated	MBE	RMSE	r
Mean air temperature (°C)	28.3	28.1	-0.24	0.36	0.996
Diurnal range (K)	7.4	7.2	-0.20	0.23	0.993
Mean RH (%)	74.0	74.0	0.00	0.00	1.000
Mean wind speed (m/s)	3.3	3.3	0.01	0.03	0.999
Rainfall (mm) [†]	1,005.0	944.2	-5.07	9.71	0.995

MBE, RMSE and r are computed over the 12 monthly means. [†]Annual totals; MBE and RMSE in mm/month.

A. Weather and Over-Water Microclimate

The synthetic year draws daily rain occurrence and amounts, a first-order autoregressive daily clearness index and daily temperature anomalies from monthly normals. Hourly GHI is the Haurwitz clear-sky value [36] scaled by the clearness index. Air temperature follows an asymmetric diurnal cycle with its minimum near 05:00 and its maximum near 14:00. The dew point T_d is held constant over each day, which reproduces the diurnal swing of RH, and is shifted per month so that the monthly mean RH matches the normal. Wind speed includes a mean-preserving afternoon sea breeze. Table III compares 20 generated years with the normals they are built from. This shows that the generator is unbiased with respect to its inputs; it is *not* an external validation, because no measured hourly series was available to this study. The RH match is exact by construction, and the monthly clearness index (hence GHI) is a calibrated input that has not been checked against measurements.

Over water, the air temperature is pulled towards the water temperature T_w , taken as the 30-day running mean of air temperature plus 0.5 K. The dew point rises by ΔT_d because of evaporation, and the wind speed increases over the smoother surface:

$$\begin{aligned} T'_a &= T_a + \alpha(T_w - T_a), & v' &= k_v v, \\ T'_d &= \min(T_d + \Delta T_d, T'_a). \end{aligned} \quad (1)$$

Relative humidity follows from the Magnus relation $e_s(T) = 6.112 \exp[17.62T/(243.12 + T)]$ hPa as $RH = 100 e_s(T'_d)/e_s(T'_a)$.

B. Irradiance, Incidence and Spectrum

Solar position uses Spencer's series [37]. GHI is decomposed with the Erbs correlation [38] and transposed to the

array plane with the Hay-Davies model [39]:

$$G = G_b \cos \theta + G_d \left[A_i R_b + (1 - A_i) \frac{1 + \cos \beta}{2} \right] + \rho G_h \frac{1 - \cos \beta}{2}, \quad (2)$$

where θ is the angle of incidence, β the tilt, A_i the anisotropy index, R_b the beam ratio and ρ the albedo (0.07 for water, 0.20 for land). Beam reflection losses use the ASHRAE modifier $IAM = 1 - b_0(1/\cos \theta - 1)$ with $b_0 = 0.05$; diffuse and ground-reflected light take a constant factor of 0.95. Humid air absorbs mainly in infrared bands where c-Si responds weakly, so the mono-Si spectral factor is computed from the Kastner-Young air mass AM [40] and the precipitable water w [41]:

$$M = c_1 + c_2 AM + c_3 w + c_4 \sqrt{AM} + c_5 \sqrt{w} + c_6 AM/\sqrt{w}, \quad (3)$$

with the coefficients of [42].

C. Module Temperature, Surface Humidity and Dew

Module temperature follows the Faiman model [43]. A net long-wave loss of $q_{sky} = 40 \text{ W/m}^2$ to the night sky applies when $G < 50 \text{ W/m}^2$:

$$T_m = T'_a + \frac{G - q_{sky} \mathcal{H}[G < 50]}{U_0 + U_1 v'}. \quad (4)$$

The land coefficients are the open-rack defaults. The baseline FPV coefficients (Table II) represent a well-ventilated pontoon design [7], [8]. Because measured values for other designs are considerably lower [11]–[14], the Monte Carlo ranges include FPV coefficients no better than those of open racks. The RH of the air layer in contact with the glass is

$$RH_s = 100 e_s(T'_d)/e_s(T_m). \quad (5)$$

Dew forms when $T_m - T'_d < 1 \text{ K}$ and persists after sunrise until $T_m - T'_d > 4 \text{ K}$. While dew is present, an optical loss of 4% is applied. This value is an engineering assumption, not a measured quantity; Section IV-C shows that the conclusions are insensitive to it over 0–6%.

D. Dust and Sea-Salt Soiling

Dust and salt surface densities evolve by a daily mass balance. Dust accumulates at r_d , and salt at $r_s v/v_0$ with $v_0 = 3 \text{ m/s}$, because sea-spray generation rises with wind. Rain above 2 mm/day removes a fraction $\eta_d f_\beta$ of the dust and $\eta_s f_\beta$ of the salt, with $\eta_d = 0.80$, $\eta_s = 0.95$ (salt is soluble) and $f_\beta = \sin \beta / \sin 20^\circ$. When the daily maximum RH exceeds the NaCl deliquescence point of 75%, dissolved salt recrystallizes as a crust that cements dust to the glass [29]; on such days the dust-cleaning efficiency is reduced by the cementation factor (baseline 0.5). All plants are cleaned every 15 days with 95% efficiency. The soiling ratio is

$$SR = \max(SR_{\min}, 1 - k_d m_d - k_s m_s), \quad (6)$$

with $k_d = 0.035$ and $k_s = 0.060 \text{ m}^2/\text{g}$. The deposition rates and coefficients are not site-measured, so they are varied in Sections IV-F and III-I.

E. Electrical Model

The effective irradiance G_{eff} includes the incidence, spectral, soiling and dew corrections. DC power follows the PVWatts model [44]:

$$P_{\text{dc}} = P_0 \frac{G_{\text{eff}}}{1000} [1 + \gamma(T_m - 25)] \prod_j (1 - \ell_j), \quad (7)$$

where $\gamma = -0.0035/\text{K}$ and ℓ_j are the wiring, mismatch, wave-motion and light-induced degradation (LID, 1%) losses. The PVWatts inverter curve (98% nominal) with clipping and an availability factor follow.

F. Module Degradation Model

The annual module degradation rate is the sum of four mechanisms:

$$R = R_0 + R_{\text{TH}} + R_{\text{PID}} + R_C, \quad (8)$$

where $R_0 = 0.30\%/ \text{yr}$ covers ultraviolet exposure and thermal cycling. Encapsulant moisture follows the surface humidity with a first-order lag ($\tau = 48 \text{ h}$) [16]:

$$RH_e(t) = RH_e(t-1) + \kappa [RH_s(t) - RH_e(t-1)], \quad (9)$$

with $\kappa = 1 - e^{-\Delta t/\tau}$. With the Arrhenius factor

$$a_E(T_m) = \exp \left[-\frac{E}{k_B} \left(\frac{1}{T_m} - \frac{1}{T_{\text{ref}}} \right) \right], \quad (10)$$

the Peck rate [15] averaged over the year is

$$R_{\text{TH}} = R_{\text{TH}}^{\text{ref}} \left\langle a_{E_a}(T_m) \left(\frac{\phi RH_e}{RH_{\text{ref}}} \right)^n \right\rangle, \quad (11)$$

with $E_a = 0.79 \text{ eV}$, $n = 3$, $T_{\text{ref}} = 25^\circ \text{C}$ and $RH_{\text{ref}} = 60\%$. The factor $\phi = 1$ applies to standard glass-backsheet modules. Because RH_s depends on T_m , cooling a module lowers the Arrhenius factor but raises RH_e ; (11) resolves this trade-off.

PID acts only while the array is energized ($G > 50 \text{ W/m}^2$):

$$R_{\text{PID}} = R_{\text{PID}}^{\text{ref}} (1 - \mu_{\text{PID}}) \langle a_{E_{\text{PID}}}(T_m) f_{\text{leak}} \rangle_{\text{on}}, \quad (12)$$

$$f_{\text{leak}} = \left[1 + \exp \left(-\frac{RH - (RH_{50} - \Delta_s s)}{5} \right) \right]^{-1}, \quad (13)$$

where $E_{\text{PID}} = 0.9 \text{ eV}$, $RH_{50} = 70\%$ and $s = \min(m_s/0.5, 1)$ is the salt coverage. The leakage term follows the strong humidity dependence of PID [20], [21]. A full hygroscopic salt film lowers the conduction threshold by $\Delta_s = 15\% \text{ RH}$, consistent with the orders-of-magnitude drop in glass surface resistance measured with sea salt [23], [24].

The corrosion rate follows the ISO 9223 logic of time of wetness and chloride deposition [27]:

$$R_C = R_C^{\text{ref}} \frac{TOW}{TOW_{\text{ref}}} \sqrt{\frac{D_{\text{Cl}}}{D_{\text{ref}}}} (1 - \mu_C), \quad (14)$$

where TOW counts the hours with $RH > 80\%$ or dew on the module, $TOW_{\text{ref}} = 2500 \text{ h}$ and $D_{\text{ref}} = 60 \text{ mg}/(\text{m}^2 \text{ d})$.

The reference rates ($R_{\text{TH}}^{\text{ref}} = 0.10$, $R_{\text{PID}}^{\text{ref}} = 0.06$ and $R_C^{\text{ref}} = 0.10\%/ \text{yr}$) were calibrated so that standard modules on land at this site degrade by roughly 0.9–1%/yr. That is the order of magnitude reported for hot-humid Indian climates [26], above the global median of about 0.5%/yr [25]. They are calibration constants, not measured site values, and are varied over a factor of about two in the Monte Carlo.

TABLE IV
MITIGATION OPTIONS: QUALIFICATION STANDARDS AND MODEL ASSUMPTIONS

Option	Qualification	Model representation	Assumed value (MC range)
PID-resistant cell/ encapsulant	IEC TS 62804-1:2025	Factor $(1 - \mu_{\text{PID}})$ on R_{PID}	$\mu_{\text{PID}} = 0.9$ (0.5–0.95)
Salt-mist-hardened module	IEC 61701:2020	Factor $(1 - \mu_C)$ on R_C	$\mu_C = 0.5$ (0.2–0.7)
Glass-glass, low-permeability encapsulant	No specific standard	Factor ϕ on RH_e	$\phi = 0.8$
More frequent cleaning	O&M practice	Cleaning interval	7 days (vs. 15)

Assumed cost adders: 0.5, 1.0 and 2.0 INR/Wp for the three module options and 100 INR/kWp/yr per additional fortnightly cleaning.

G. Mitigation Options and Their Assumed Effects

Qualification standards define test procedures; they do not establish a numerical reduction of field degradation. The mitigation cases are therefore described by the technology and the standard used to qualify it, and their effect enters the model through *explicitly assumed* factors (Table IV). Salt-mist resistance is qualified with IEC 61701 [28]. PID susceptibility of crystalline-silicon modules is evaluated with IEC TS 62804-1 [45]; accelerated PID tests have been used to predict field PID rates [46]. The factors μ_{PID} and μ_C are sampled over wide ranges in Section III-I. In the rest of the paper, “(M)” denotes modules with both PID-resistant technology and salt-mist hardening.

H. Lifetime Energy and Economics

Year- y energy is $E_y = E_1(1 - R/100)^{y-1}$ over $N = 25$ years, and the LCOE is

$$\text{LCOE} = \frac{C_0 + \sum_{y=1}^N O_y(1+r)^{-y} - \sum_{y=1}^N B_y(1+r)^{-y}}{\sum_{y=1}^N E_y(1+r)^{-y}}, \quad (15)$$

with a discount rate $r = 9\%$ and O&M costs O_y escalating by 5%/yr. $B_y = c_w V_w$ is an optional water benefit: for the freshwater plant, V_w is the evaporation avoided by shading ($10 \text{ m}^2/\text{kWp}$, 65% suppression of 1800 mm/yr pan evaporation) and c_w a water value. The costs are illustrative Indian-market values. Because LCOE is linear in C_0 and c_w , break-even values follow in closed form.

I. Uncertainty Quantification

Two sources of uncertainty are treated separately. *Weather variability* is represented by 20 independent synthetic years. *Parameter uncertainty* is propagated by Monte Carlo sampling of 26 parameters (1,000 samples): the three degradation reference rates, the dew optical loss (0–6%), the soiling coefficients and cementation factor, the land and FPV heat-loss coefficients (U_0 : 22–29 and 22–45 W/(m² K); U_1 : 4.5–8 and 2.5–10 W s/(m³ K)), the wind factor, the wave-motion mismatch, α and ΔT_d for each FPV site, the dust, salt and chloride deposition (log-uniform, $0.5 \times -2 \times$), μ_{PID} , μ_C and the CAPEX of each plant. Parameters shared by the plants (e.g., the reference rates) take the same draw in all scenarios of a

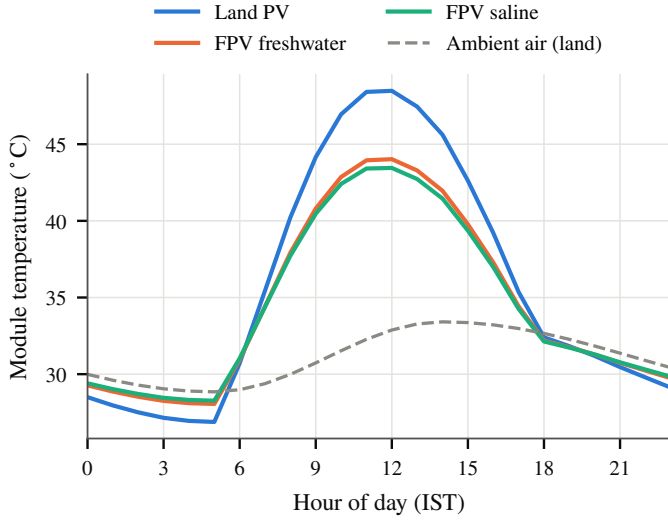


Fig. 3. Mean module temperature by hour of day in May for the three plants, with the site air temperature (baseline parameters).

TABLE V
YEAR-1 PERFORMANCE (BASELINE)

Metric	Land	FPV fresh	FPV saline
POA irradiation (kW h/m ²)	1963	1953	1953
Specific yield (kWh/kWp)	1608	1634	1590
Performance ratio	0.819	0.837	0.814
CUF, DC basis (%)	18.35	18.66	18.15
Mean daytime T_m (°C)	39.5	36.5	36.2
Maximum T_m (°C)	57.3	50.0	49.4
Mean $T_m - T_a$, daytime (K)	10.1	7.3	7.0
Mean air RH at array (%)	74.0	77.5	79.7
Dew on glass (h/yr)	1,163	1,092	1,389
Mean daytime soiling ratio	0.965	0.988	0.969

sample, so the scenario differences are paired. Spearman rank correlations identify the dominant parameters. The ranges, seeds and baseline values are stored in a machine-readable file in the repository.

IV. RESULTS

Unless stated otherwise, the results below are model predictions under the baseline assumptions of Table II.

A. Thermal Behavior

Fig. 3 shows the mean diurnal module temperature in May. Over the year, floating modules run 3.0 K cooler in daytime and their peak temperature is 7.3 K lower (Table V). The daytime module-to-air rise falls from 10.1 K on land to 7.3 K (freshwater) and 7.0 K (saline). At night the air over water stays warmer and the modules follow it, while the higher dew point keeps the air near the array more humid. Dew is present for 1,163, 1,092 and 1,389 h/yr at the land, freshwater and saline plants. The thermal advantage depends directly on the assumed FPV heat-loss coefficients, which are the most uncertain thermal inputs (Section IV-H).

TABLE VI
YEAR-1 LOSS BREAKDOWN (BASELINE). PERCENTAGES ARE SEQUENTIAL LOSSES RELATIVE TO THE PRECEDING STAGE AND ARE NOT DIRECTLY ADDITIVE.

Loss stage (%)	Land	FPV fresh	FPV saline
Incidence angle (IAM)	3.44	3.47	3.47
Spectral (humidity)	-0.23	-0.24	-0.25
Soiling (dust + salt)	3.67	1.27	3.29
Dew / condensation	0.02	0.04	0.05
Module temperature	6.12	4.78	4.64
DC (wiring, mismatch, motion, LID)	3.46	4.62	4.91
Inverter and clipping	2.03	2.03	2.03
Availability	1.00	1.50	2.00

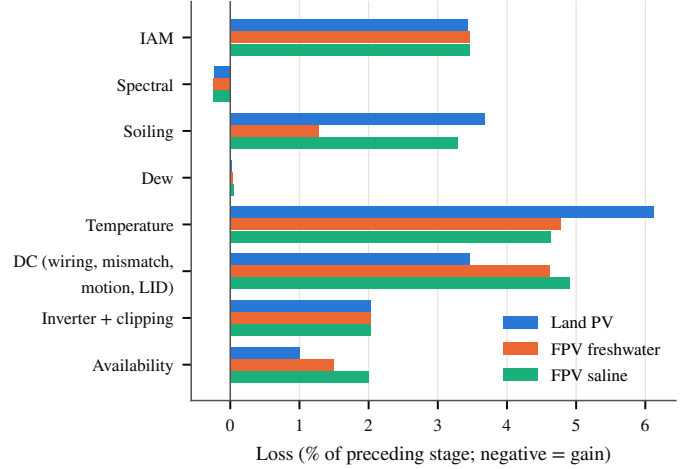


Fig. 4. Year-1 sequential loss breakdown (baseline). Negative values are gains.

B. Energy Yield and Loss Breakdown

Freshwater FPV gives the highest year-1 specific yield, 1634 kWh/kWp, +1.6% relative to land (PR 0.837 against 0.819). Table VI and Fig. 4 show where the difference comes from. The temperature loss falls from 6.12% to 4.78%, and the soiling loss falls from 3.67% to 1.27% because little dust reaches the water. Part of the gain is lost to higher DC losses (cabling and wave-induced mismatch: 4.62% against 3.46%), lower availability and a slightly lower POA irradiation. Salt soiling brings the saline plant's soiling loss back to 3.29%, and its year-1 yield is 1590 kWh/kWp (−1.1%). Humidity gives a small spectral gain of about 0.24% at all sites, and dew costs less than 0.1%.

The FPV advantage in monthly PR (Fig. 5) is largest in the dry, clear months. The saline plant performs worst in the pre-monsoon months, when strong onshore winds bring salt and little rain falls. Between cleanings, salt deposition gives the saline plant a saw-tooth soiling pattern similar to dust soiling on land (Fig. 6).

C. Sensitivity to Dew Optical Loss

Varying the dew optical loss from 0 to 6% changes the year-1 yield difference relative to land by less than 0.05 percentage point: it ranges from 1.64% to 1.66% for freshwater FPV and from −1.10% to −1.06% for saline FPV. Dew matters through time of wetness, not as an optical loss.

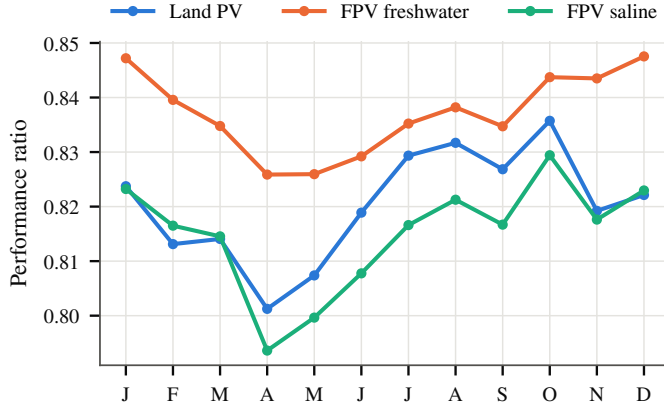


Fig. 5. Monthly performance ratio (baseline).

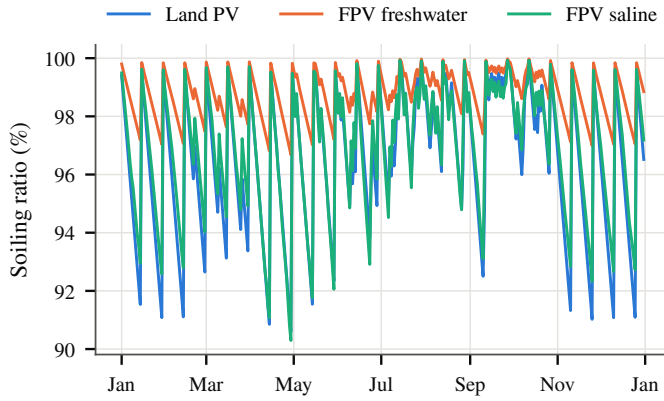


Fig. 6. Daily soiling ratio with rain cleaning, NaCl deliquescence and 15-day manual cleaning (baseline).

D. Module Degradation

Table VII and Fig. 7 resolve the modelled degradation by mechanism. Floating modules are cooler, but their encapsulant equilibrates at a higher moisture level: mean RH_e is 63.5%, 68.4% and 70.4% at the land, freshwater and saline plants. Through the cubic humidity term of (11), this cancels the Arrhenius benefit of cooling. R_{TH} is 0.34, 0.34 and 0.36%/yr. In this model, therefore, *cooling by water does not translate into slower moisture-driven ageing at a humid site*. PID is predicted at 0.15, 0.15 and 0.21%/yr; the saline value is highest because the salt film makes the glass conductive at lower RH. Corrosion separates the sites most clearly. Time of wetness rises from 3,055 h/yr on land to 3,719 and 4,288 h/yr over water, and with 180 mg/(m² d) of chloride, R_C reaches 0.30%/yr at the saline plant (0.09%/yr on land). Under the baseline assumptions, total degradation is 0.88, 0.84 and 1.16%/yr.

E. Lifetime Energy and Mitigation

Over 25 years (Fig. 8), freshwater FPV delivers 37,004 MWh/MWp (+2.1% against land) and saline FPV 34,677 MWh/MWp (−4.3%). Table VIII evaluates the mitigation options of Table IV for the saline plant one at a time. PID-resistant modules alone recover 2.2% of lifetime energy,

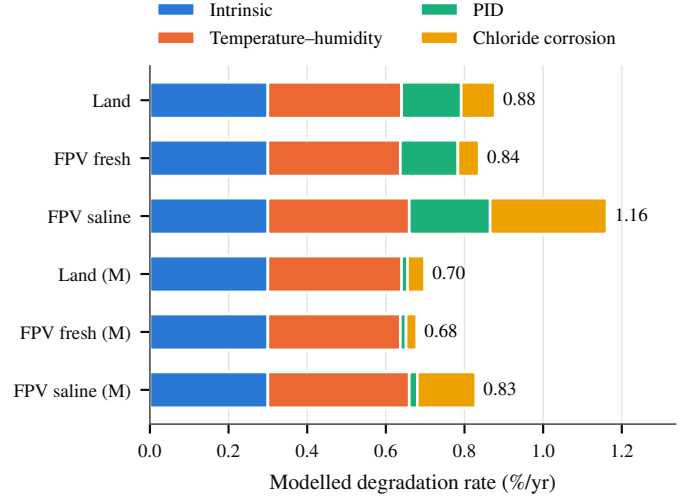


Fig. 7. Modelled module degradation rate by mechanism (baseline). (M): PID-resistant and salt-mist-hardened modules with the assumed factors of Table IV.

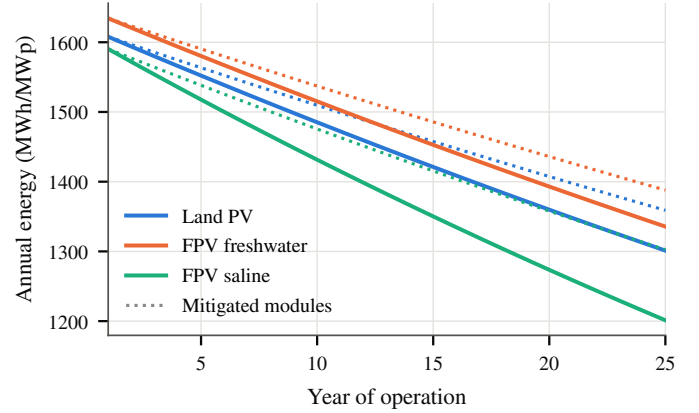


Fig. 8. Annual energy over 25 years (baseline). Dotted lines: PID-resistant and salt-mist-hardened modules (M).

salt-mist hardening alone 1.7%, and both together 3.9%, still −0.5% below land. Only when a lower-permeability glass–glass package is added does the saline plant exceed the land plant (1.5%); with a 7-day cleaning interval as well, it reaches 3.3%. These recoveries follow directly from the assumed factors and should be read as what-if results.

F. Thresholds for Salinity and Humidity

Fig. 9(a) varies the salt-soiling deposition of the saline plant with the chloride load held at baseline. Saline FPV out-yields land PV in year 1 only below about 35 mg/(m² d) of salt deposition. Its 25-year energy stays below land at every salt level, and is still −0.9% at zero salt deposition. Varying the chloride deposition with salt soiling at baseline gives the same picture: the degradation rate rises from 0.87%/yr at zero chloride to 1.31%/yr at 400 mg/(m² d), and no chloride level in 0–400 restores lifetime parity (at zero chloride the deficit is −0.9%). Under the baseline assumptions, salt soiling, salt-assisted PID and the extra DC and availability losses each contribute to the saline deficit, so no single mitigation of

TABLE VII
MODELLED MODULE DEGRADATION, LIFETIME ENERGY AND LCOE (BASELINE, PER MWp; M = PID-RESISTANT AND SALT-MIST-HARDENED MODULES WITH THE ASSUMED FACTORS OF TABLE IV)

	Land	FPV fresh	FPV saline	Land (M)	FPV fresh (M)	FPV saline (M)
R_0 intrinsic (%/yr)	0.30	0.30	0.30	0.30	0.30	0.30
R_{TH} temperature-humidity (%/yr)	0.34	0.34	0.36	0.34	0.34	0.36
R_{PID} (%/yr)	0.15	0.15	0.21	0.02	0.01	0.02
R_C chloride corrosion (%/yr)	0.09	0.05	0.30	0.04	0.03	0.15
Total R (%/yr)	0.88	0.84	1.16	0.70	0.68	0.83
Time of wetness (h/yr)	3,055	3,719	4,288	3,055	3,719	4,288
Mean encapsulant RH (%)	63.5	68.4	70.4	63.5	68.4	70.4
Module capacity in year 25 (%)	80.9	81.7	75.5	84.5	84.9	81.9
25-year energy (MWh)	36,233	37,004	34,677	37,002	37,698	36,045
LCOE (INR/kWh)	2.83	3.54	4.43	2.90	3.60	4.43

TABLE VIII
MITIGATION OPTIONS FOR THE SALINE FPV PLANT (ASSUMED EFFECTS OF TABLE IV; PER MWp)

Option	CAPEX (INR/Wp)	R (%/yr)	25-yr energy (MWh)	Recovery vs. standard (%)	vs. land PV (%)	LCOE (INR/kWh)
Standard module	52.0	1.16	34,677	0.0	-4.3	4.43
PID-resistant	52.5	0.98	35,427	2.2	-2.2	4.41
Salt-mist-hardened	53.0	1.01	35,277	1.7	-2.6	4.45
PID + salt-mist	53.5	0.83	36,045	3.9	-0.5	4.43
PID + salt-mist + glass-glass	55.5	0.65	36,792	6.1	1.5	4.50
PID + salt-mist + glass-glass + 7-day cleaning	55.5	0.65	37,421	7.9	3.3	4.54

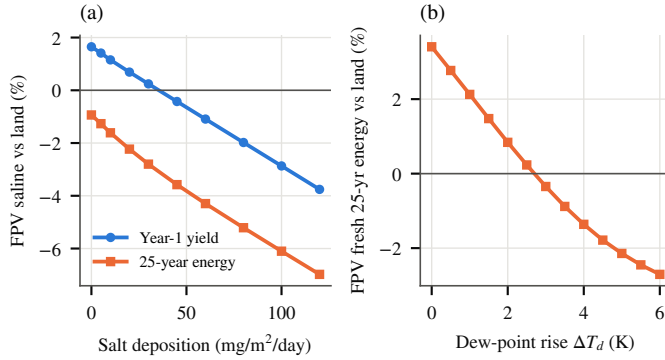


Fig. 9. Break-even thresholds relative to land PV. (a) Saline FPV versus salt-soiling deposition (chloride at baseline). (b) Freshwater FPV 25-year energy versus dew-point rise over water.

salinity closes it. For freshwater FPV, Fig. 9(b) varies the dew-point rise over water. The lifetime advantage over land vanishes at $\Delta T_d^* \approx 2.7$ K, where the time of wetness reaches about 5,745 h/yr. Measuring the humidity increment over a reservoir is therefore a priority for site assessment.

G. Joint Sensitivity to Humidity and Salinity

Fig. 10 varies the dew-point rise over water and scales salt and chloride deposition together for the saline plant. The degradation rate ranges from 0.78 to 1.49%/yr and the 25-year energy from 32,498 to 37,005 MWh/MWp. Each kelvin of dew-point rise adds about 0.167%/yr at baseline salinity. Across the ranges swept, salinity moves lifetime energy by about 2,854 MWh/MWp and moisture by about 1,674 MWh/MWp.

TABLE IX
DIFFERENCES FROM LAND PV OVER 20 WEATHER REALIZATIONS, BASELINE PARAMETERS (MEAN $\pm$ SD)

	FPV fresh	FPV saline
Year-1 yield (%)	$+1.71 \pm 0.08$	-1.08 ± 0.05
25-year energy (%)	$+2.17 \pm 0.10$	-4.32 ± 0.06
Degradation rate (%/yr)	-0.039 ± 0.003	$+0.288 \pm 0.002$

H. Weather Variability Versus Parameter Uncertainty

Weather variability. Over 20 synthetic years (Table IX, Fig. 11), the land yield varies between 1,570 and 1,649 kWh/kWp, but the differences between plants are stable, with standard deviations of about 0.1 percentage point. This is stochastic robustness conditional on the weather generator, not empirical validation.

Parameter uncertainty is much larger (Table X, Fig. 12). The 95% interval of the freshwater FPV lifetime-energy difference is $[-3.6, 6.4]\%$, positive in only 73% of samples. For saline FPV it is $[-11.2, 0.7]\%$, negative in all but 4% of samples; with mitigated modules it is $[-7.4, 3.6]\%$. For the saline difference, the rank correlations (Fig. 12b) show that the land dust deposition and the FPV heat-loss coefficients dominate, followed by the saline salt deposition; the corrosion reference rate, the land U_1 and the chloride deposition come next. For the freshwater difference the leading parameters are Land dust deposition ($\rho_s = 0.57$), FPV U_1 (0.49) and FPV U_0 (0.39). The freshwater FPV advantage therefore rests mainly on the floating design's cooling and on how dusty the land alternative is, while the saline deficit persists over most of the plausible parameter space.

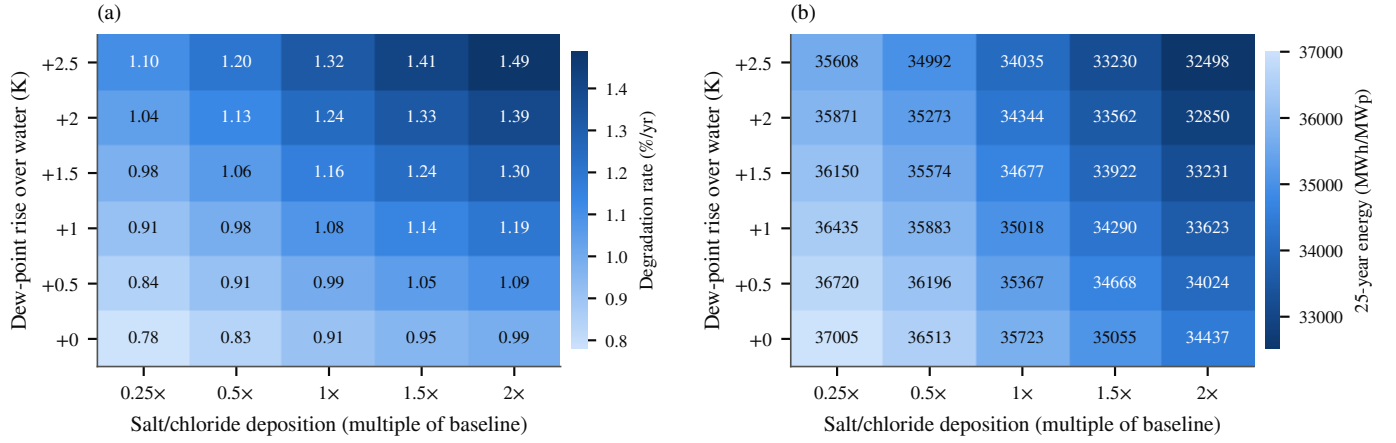


Fig. 10. Saline FPV plant versus dew-point rise over water and salt/chloride load: (a) total modelled degradation rate and (b) 25-year energy.

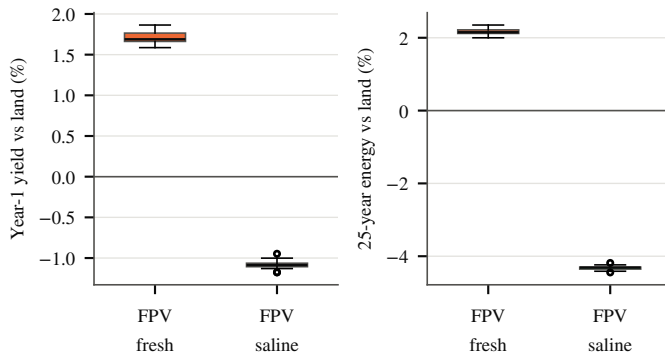


Fig. 11. Weather variability only: distribution over 20 synthetic years of the FPV year-1 yield and 25-year energy relative to land PV.

TABLE X
PARAMETER UNCERTAINTY: MONTE CARLO OVER 26 PARAMETERS
(1,000 SAMPLES, BASELINE WEATHER)

Output	Median	95% interval	$P(> 0)$ (%)
Year-1 yield, land (kWh/kWp)	1,605	[1,542, 1,638]	
Year-1 yield, FPV fresh (kWh/kWp)	1,623	[1,589, 1,645]	
Year-1 yield, FPV saline (kWh/kWp)	1,581	[1,517, 1,617]	
Degradation, land (%/yr)	0.89	[0.71, 1.11]	
Degradation, FPV fresh (%/yr)	0.88	[0.68, 1.15]	
Degradation, FPV saline (%/yr)	1.21	[0.89, 1.63]	
Degradation, FPV saline (M) (%/yr)	0.90	[0.68, 1.21]	
Δ yield, fresh — land (%)	1.11	[−1.84, 5.28]	76
Δ yield, saline — land (%)	−1.40	[−6.06, 2.93]	25
Δ 25-yr energy, fresh — land (%)	1.33	[−3.64, 6.41]	73
Δ 25-yr energy, saline — land (%)	−4.94	[−11.23, 0.65]	4
Δ 25-yr energy, saline (M) — land (%)	−1.62	[−7.35, 3.62]	27
LCOE, land (INR/kWh)	2.84	[2.50, 3.20]	
LCOE, FPV fresh (INR/kWh)	3.60	[3.13, 4.06]	
LCOE, FPV saline (INR/kWh)	4.51	[3.91, 5.12]	

I. Economics and Break-Even Values

Under the assumed costs, FPV is more expensive per kWh: 3.54 INR/kWh (freshwater) and 4.43 INR/kWh (saline) against 2.83 INR/kWh on land. In the Monte Carlo, freshwater FPV is more expensive than land PV in 99.2% of samples and saline FPV in 100.0%. Solving (15) for parity with land PV gives the following break-even values:

- CAPEX: 34.3 INR/Wp for freshwater FPV (baseline 45)

and 29.1 INR/Wp for saline FPV (baseline 52).

- *Land cost*: land PV would need a further 10.5 INR/Wp of land-related cost to match freshwater FPV, and 23.6 INR/Wp to match saline FPV.
- *Water value*: valuing the avoided evaporation (11,700 m³ per MWp and year) lowers the freshwater LCOE to 3.47, 3.39 and 3.24 INR/kWh at 10, 20 and 40 INR/m³. Parity needs about 93 INR/m³.

FPV is thus justified mainly where land is scarce or costly, where reservoir water has a high value, or where FPV CAPEX falls substantially. No salinity level makes saline FPV cost-competitive at the assumed CAPEX.

V. DISCUSSION

A. Comparison With Previous Work

The baseline thermal advantage of FPV (about 3 K in day-time, 7 K at peak) lies within the range reported for open, well-ventilated designs [7], [8]. It exceeds the advantage measured for less ventilated systems [11], [14]. The Monte Carlo makes this explicit: with FPV heat-loss coefficients at the low end of the measured range, the freshwater FPV advantage disappears. The predicted salt-driven increase in degradation is larger than the 0.05–0.13 percentage-point effect of salt reported for offshore modules [32]. The difference arises because this work also includes salt soiling, salt-assisted PID and chloride corrosion, whose calibration remains uncertain. Mannino *et al.* [31] found that degradation models need coefficients calibrated to the installation environment; that conclusion applies equally to the reference rates used here.

B. Implications for Design and Siting

The analysis separates two questions that are often merged: whether FPV produces more energy in year 1, and whether it produces more over its life. For freshwater reservoirs, the baseline answer is yes to both, but the answer depends mainly on the float design's heat-loss coefficients and on the humidity increment over the water. For saline sites, both baseline answers are no, and the lifetime deficit holds over most of the parameter space. These model-based findings suggest the following:

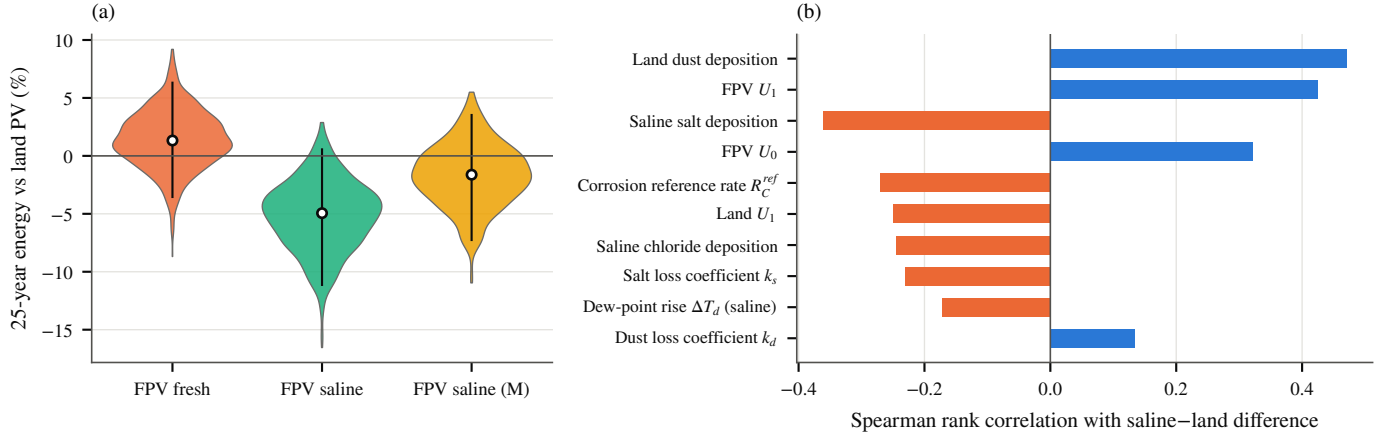


Fig. 12. Parameter uncertainty (1,000 Monte Carlo samples). (a) 25-year energy relative to land PV; dots mark the median and bars the 95% interval. (b) The ten parameters with the largest Spearman rank correlation with the saline minus land difference in 25-year energy.

- Verify the heat-loss coefficients of the chosen float design and measure RH and dew point over the water before relying on a lifetime gain.
- At saline or near-shore sites, select PID-resistant module technology (qualified with IEC TS 62804-1) and modules qualified for salt mist (IEC 61701). Consider low-permeability glass-glass packages, and treat the size of their field benefit as uncertain until field data exist.
- Use negative-pole grounding or night-time PID recovery, marine-grade connectors and corrosion-resistant structures designed for the ISO 9223 corrosivity class of the site.
- At saline sites, shorten the cleaning interval in the dry, windy pre-monsoon season, using demineralized water.
- Monitor module temperature, RH at the array, insulation resistance and soiling in line with IEC 61724-1 [47], so that the parameters can be recalibrated with field data.

C. Limitations

- *Validation.* The weather is a synthetic year that reproduces its input normals (Table III) but has not been checked against a measured hourly series. No FPV module-temperature or performance data were available to validate the thermal, soiling or degradation models.
- *Assumed parameters.* The over-water microclimate, the deposition rates, the degradation reference rates and the mitigation factors are assumptions. Their uncertainty is propagated, but the ranges themselves are judgements.
- *Degradation physics.* The model is linear in time and additive across mechanisms. It does not represent PID saturation or recovery, interactions between mechanisms, abrupt failures of corroded interconnects, or the higher operating temperature of glass-glass modules.
- *System scope.* Float and mooring fatigue, structural corrosion [33], [34], storm damage, bird soiling and bio-fouling are not modeled.

The framework accepts measured hourly weather, so the next step is to calibrate it against monitoring data from operating FPV plants in coastal Andhra Pradesh.

VI. CONCLUSION

A coupled thermal, soiling and module-degradation model was used to compare land-mounted, freshwater floating and saline floating PV for a hot, humid, coastal site in India. Under the baseline assumptions, water cooling lowers daytime module temperature by 3.0 K and gives freshwater FPV +1.6% year-1 yield and +2.1% lifetime energy. The same water body raises the humidity at the modules, and in the model this cancels the expected temperature benefit for moisture-driven degradation. At the saline site, chloride corrosion and salt-assisted PID raise the predicted degradation rate to 1.16%/yr, and lifetime energy falls by -4.3%. Parameter uncertainty is much larger than weather variability. The saline lifetime deficit is robust over most of the plausible parameter space, whereas the freshwater advantage depends on the floating design's heat-loss coefficients and on the humidity increment over the water, and vanishes above a dew-point rise of about 2.7 K. Under the baseline parameterization, salinity is the dominant driver of the predicted lifetime penalty. Module selection for PID and salt-mist resistance is the most effective modelled mitigation, although its size rests on assumed reduction factors. These conclusions are model-based and call for field calibration. Assessments of FPV in humid and saline regions should report lifetime energy with mechanism-resolved degradation and quantified uncertainty, not first-year yield alone.

DATA AVAILABILITY

The open-source Python implementation (fpv_analysis), its unit tests, the scenario parameters and Monte Carlo ranges in machine-readable form (paper/generated/parameters.json), all random seeds, the pinned package versions and every generated table are available in the project repository <https://github.com/subrahmanyamtanala-rgb/floating-solar-photovoltaic-system-for-humid-and-saline-conditions>. The single command `./reproduce.sh` regenerates every number, table and figure in this paper.

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